Intelligent Reasoning System Project

Project title: DC-Guardian: Data Center Physical & Cybersecurity Correlated with Predictive Maintenance

Group Number: Practice Module **Group 22**

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Introduction

Data centers are critical infrastructures that support essential digital services, business operations, and information systems. Maintaining their security, safety, and operational reliability requires continuous monitoring of both physical and cybersecurity threats, as well as environmental conditions that may affect equipment performance. However, these aspects are often managed through separate tools and systems, making it difficult to identify relationships between events and recognize potential risks at an early stage. For example, an unauthorized individual entering a data center, failing to wear the required Personal Protective Equipment (PPE), abnormal humidity or smoke levels, and repeated unauthorized SSH login attempts may appear as unrelated alerts when monitored independently, even though they may contribute to a broader security, safety, or maintenance incident.

DC-Guardian is an intelligent, unified data center protection and maintenance monitoring system that integrates physical security, environmental monitoring, cybersecurity, and maintenance-related intelligence into a single interactive dashboard. The system combines facial recognition to identify authorized personnel, PPE detection to monitor safety compliance, IoT sensors to monitor environmental hazards such as humidity and smoke, and cybersecurity anomaly detection to identify suspicious activities such as unauthorized or brute-force SSH login attempts. In addition, the system can support maintenance activities by identifying abnormal environmental conditions and correlating them with relevant equipment or maintenance information.

The project is relevant because modern data centers face increasingly complex and interconnected security, safety, and operational risks. Detecting individual events alone may not provide sufficient understanding of the overall situation. By correlating information from multiple sources, DC-Guardian aims to provide security and maintenance personnel with a comprehensive view of incidents, identify potential risks before they escalate, reduce isolated or redundant alerts, and support faster and more informed decision-making. The integration of Artificial Intelligence, IoT, Knowledge Graphs, and RAG demonstrates how emerging technologies can be applied to improve intelligent data centers protection and proactive maintenance.

The main goals of DC-Guardian are to develop a unified platform for monitoring physical security, environmental conditions, cybersecurity, and maintenance-related risks, accurately identify authorized and unauthorized personnel and detect PPE compliance, monitor environmental conditions and identify abnormal humidity and smoke events that may require maintenance or intervention, detect suspicious and unauthorized SSH login activities, use Multi-Agent RAG and a Neo4j Knowledge Graph to correlate security, environmental, and maintenance-related events across-domains, and provide unified incident triage, maintenance insights, explanations, and actionable recommendations through an interactive dashboard. Ultimately, DC-Guardian aims to demonstrate a proactive and intelligent approach to data centers protection and maintenance by moving beyond isolated monitoring toward integrated, cross-domain security and operational awareness.

Project Background / Market Context

Data centers have become a fundamental part of modern digital infrastructure, supporting cloud computing, enterprise applications, artificial intelligence, financial services, telecommunications, and other mission-critical services. As organizations become increasingly dependent on data center availability and reliability, protecting these facilities requires more than traditional physical access control or cybersecurity tools. Modern data centers must address physical security, environmental hazards, cybersecurity threats, operational reliability, and maintenance simultaneously. The growing scale and complexity of data center infrastructure is also increasing the need for continuous monitoring and intelligent security solutions. Current market research indicates continued growth in the data center physical security market, driven by the expansion of data center infrastructure, increasing cyber-physical risks, and the adoption of AI-enabled monitoring technologies.

A major challenge addressed by DC-Guardian is the fragmentation of existing monitoring and security systems. Data centers commonly rely on different technologies for access control, video surveillance, environmental monitoring, equipment monitoring, and cybersecurity. Although these systems can detect individual events, they may operate independently and provide limited understanding of how events across different domains are related.

For example, a facial recognition system may identify an unauthorized individual, while an environmental monitoring system separately detects abnormal humidity or smoke, and a cybersecurity system detects repeated failed SSH login attempts. When these events are viewed independently, security or maintenance personnel may treat them as separate alerts. However, when correlated, they could indicate a much more significant incident involving physical intrusion, equipment risk, or a coordinated cyber-physical attack.

This fragmentation can lead to alert overload, delayed response, difficulty identifying the root cause of incidents, and limited situational awareness. It can also make proactive maintenance more difficult because abnormal environmental or operational conditions may not be connected to the relevant equipment or maintenance information.

The current market is moving toward AI-enabled, integrated, and centralized security platforms. Research on data center physical security highlights the increasing use of AI and machine learning for intelligent video analytics, predictive threat detection, automated anomaly recognition, and real-time monitoring. There is also increasing convergence between physical security and IoT-enabled environmental monitoring, allowing multiple security layers to be managed through centralized platforms (Ltd, n.d.-b).

Another market trend is the movement toward cyber-physical security convergence. Recent industry research identifies the integration of biometric authentication, AI-powered video analytics, IoT sensors, environmental monitoring, and cybersecurity technologies as an emerging direction for data center protection. Predictive maintenance and continuous monitoring are also becoming increasingly important as operators seek to prevent failures and maintain service availability (Data Center Physical Security Market Growth, Trends & Industry Forecast 2034, n.d.).

From a research perspective, technologies such as Knowledge Graphs, Retrieval-Augmented Generation (RAG), multi-agent AI systems, computer vision, and anomaly detection provide opportunities to move beyond conventional alert-based monitoring. Knowledge Graphs can represent relationships between people, devices, sensors, systems, locations, and incidents, while RAG can provide agents with relevant operational procedures, maintenance documentation, and safety guidelines when analyzing an event. Multi-agent architectures can then allow specialized agents to analyze different domains and collaborate on a unified incident assessment.

Therefore, DC-Guardian fits within the broader market transition from isolated security solutions toward integrated, AI-driven cyber-physical security platforms. Its distinctive focus is on combining physical security, environmental monitoring, cybersecurity, and maintenance within one intelligent framework, while using Multi-Agent RAG and Neo4j Knowledge Graph technology to correlate cross-domain events and support unified incident triage and proactive maintenance. This approach aims to improve situational awareness, reduce fragmented alerts, and support faster and more informed responses to complex data center incidents.

Literature Review / Market Research

Literature Review

Ahmed et al. tested the predictive maintenance in data centers; the findings are that the operational and financial impact of failures is impacting computational resources and power supply equipment. They propose analysis of component failure and fixing response to support faster maintenance actions, extending equipment life by reducing downtime. These findings provide the first academic foundation for DC-Guardian, which expands on this approach by building the combination of equipment health monitoring systems with cybersecurity and access control events rather than managing maintenance as standalone functions (Ahmed et al., 2021).

What makes anomaly detection central to the project is because unauthorized access, equipment failure, and cybersecurity incidents are uncommon in comparison with regular operations. Liu, Ting and Zhou proposed Isolation Forest, which detects anomalies through isolating uncommon observations by randomly constructed trees rather than generating a full model of regular behaviors. Its efficient use of computational and memory resources make it well suited for analyzing continuous telemetry data as a baseline In DC-Guardian, Isolation Forest will be used as a baseline for identifying unusual combinations of variables such as temperature, rotational speed, torque, and tool wear. (Liu et al., 2008).

The project uses AI4I 2020 Predictive Maintenance Dataset for the initial telemetry experiment. This dataset has 10K synthetic records capturing real industrial operating factors, such as multiple machine failure modes, which is suitable for developing and testing a proof of concept (PoC) anomaly model. However, it describes an industrial machine instead of an actual data center infrastructure. As a result, it will be presented as prototype evidence, with real data center telemetry determined as a need for future assessment (Matzka, 2020).

Knowledge Graph research shows the second major foundation. Surveys of cyber–Knowledge Graphs provide that graph-based representation could integrate heterogeneous data, improve operational visibility, expose relationships and attack paths, and enable reasoning over connected entities. This supports DC-Guardian's adoption of Neo4j to capture infrastructure relationships (Sikos, 2023).

The Knowledge Graph enables DC-Guardian to answer relational questions that cannot be addressed using isolated alerts. As an example, it may identify if a thermal anomaly, failed badge attempt and unusual SSH activity are linked with the same rack or server. This allows incident priority to consider asset relationships and possible cascading effects rather than relying solely on the severity of each individual alert.

Retrieval-Augmented Generation (RAG) helps to ensure that the project's recommendations are built on relevant operational data. Lewis et al. (2020) describe RAG as an approach that combines a language model with information retrieved from external knowledge sources. In DC-Guardian, documents like Standard Operating Procedures, troubleshooting guides and incident-response plans are stored in a vector database. The Operations Agent retrieves the most relevant sections from these documents before recommending a decision. This makes its recommendations easier to test and reduces the possibility of showing answers that are not supported by the available procedures (Lewis et al., 2020).

Overall, DC-Guardian does not aim to introduce a new anomaly-detection algorithm. Its academic contribution lies in integrating and evaluating multiple intelligent reasoning techniques within a unified physical-cyber data-center use case. The project investigates whether combining model-generated events, infrastructure relationships, security context, and retrieved procedural knowledge can improve incident correlation, prioritization, and response recommendations compared with isolated single-domain monitoring.

Market Research

Although the current market shows mature products for data center infrastructure monitoring, predictive maintenance and security operations, these capabilities are mostly offered by different systems, creating an opportunity for further reasoning layers that could link their outputs.

Table 1 - Market Research

Market Research	**Primary Focus**	**DC-Guardian**
EcoStruxure IT data center infrastructure management	Focuses on the nature of vendor DCIM capabilities, real-time visibility of power cooling and IT infrastructure, managing alarms, and overall	Agrees on the demand of centralized and proactive data center monitoring. DC-Guardian extends on adding Knowledge Graph, RAG, and multi-agent correlation among operational and cybersecurity incidents.

	environmental monitoring.	
Trellis platform data center monitoring solution	Centralizing the monitoring of critical infrastructure, environmental conditions, power systems and facility alarms.	offers comparable infrastructure monitoring environment, while DC-Guardian emphasis on reasoning among telemetry, physical access and cybersecurity data.
Asset performance management software	Focuses on asset health monitoring, proactive maintenance, work management and performance management supported by AI.	Illustrates the demand for condition based/predictive maintenance. DC-Guardian is different by focusing on cross-domain data center incidents.
Splunk Enterprise Security/ Microsoft Sentinel	Focuses on cybersecurity threat detection, log collection and incident investigation.	DC-Guardian differs by providing reason over cooling, equipment health and physical-location relationships. DC-Guardian differs by making the correlation of cybersecurity events with physical location, environmental conditions, and equipment-health context a primary focus of its proposed reasoning architecture.
Security Center Unified Security Platform	Have a centralized system for access control, video surveillance, communication, and physical security event management.	DC-Guardian extends the physical-security perspective by incorporating cybersecurity, equipment/environmental monitoring, Knowledge Graph relationships, and knowledge-grounded reasoning within the proposed prototype.

The current market of data center physical security is expected to expand continuously over the years, because the organizations keep investing in access control, monitoring, and protection for increasingly critical data center facilities. This growth demonstrates the need for more effective physical security measures, alongside cybersecurity and infrastructure monitoring (Data Center Physical Security Market Growth, Trends & Industry Forecast 2034, n.d.).

Project Scope

DC-Guardian is designed as a unified cyber-physical security and predictive maintenance platform tailored for modern data center infrastructures. The project addresses the industry's move toward AI-enabled, centralized security platforms that bridge the gap between fragmented monitoring systems, and targets the growing demand for cyber-physical security convergence, in which physical access, environmental health, and digital security are managed under a single dashboard to reduce the risk of complex, multi-vector incidents.

Academically, the project explores the integration of multimodal data — ranging from IoT sensor telemetry and system logs to visual PPE compliance and facial recognition — into a cohesive reasoning framework. It adopts a hybrid data strategy, combining public benchmark datasets with synthetic infrastructure mapping to simulate a complex, real-world data center environment.

System Coverage

The following table defines the domains and capabilities that fall within the boundaries of the DC-Guardian system.

Table 2 - Domain and Coverage

Domain	**Coverage**
Physical security and Safety through CCTV	Facial recognition for authorized/unauthorized personnel identification; PPE compliance detection (hard hats, vests, gloves, etc.) via computer vision through one camera.
Predictive Maintenance / Equipment Health	Detection of abnormal equipment-health patterns using AI4I as a proxy benchmark. The prototype evaluates anomaly/failure-detection models and maps their outputs to simulated data-center assets.
Cybersecurity	Detection of anomalous or brute-force SSH login attempts from system and server logs.
Cross-domain correlation	A Neo4j Knowledge Graph modeling relationships between people, zones, racks, servers, sensors, and events, supporting the correlation of events that may otherwise appear unrelated when viewed independently.
Knowledge-grounded reasoning	A Multi-Agent RAG layer in which specialized agents retrieve relevant SOPs, compliance manuals, and response procedures to ground incident triage in operational documentation.
Presentation layer	A unified interactive dashboard for security and maintenance personnel to view correlated incidents, explanations, and recommended actions.

Intelligent Reasoning

The project's academic contribution centers on cross-domain reasoning rather than on any single detection model. Three techniques form the core of this contribution:

Table 3 - Three Core Intelligent Reasoning

Technique	Role in DC-Guardian	Academic Contribution
Knowledge Graph (Neo4j)	Represents spatial and organizational context — zones, racks, servers, personnel, and sensors.	Enables structured, quarriable reasoning. over infrastructure relationships rather than flat event lists.
Retrieval-Augmented Generation (RAG)	Ground agent decisions in Standard Operating Procedures, compliance manuals, and response documentation.	Reduces reliance on unverified model generation by anchoring triage in retrieved operational evidence.
Specialized Agents (Agentic AI)	Coordinates domain-specific agents (physical, environmental, cyber) toward a unified assessment.	Demonstrates collaborative reasoning across agents rather than isolated, single-domain alerting.

Limitations and Constraints

The project acknowledges several constraints that affect both its research validity and its market applicability. These are summarized below.

Table 4 - Limitations Summary

Limitation	Description
No synchronized real-world dataset	No public dataset combines physical security, cybersecurity, and maintenance telemetry from a single facility, so correlation relies on synthetic temporal/spatial alignment rather than observed correlation. This constrains how far results can be generalized to real deployments.
Synthetic infrastructure topology	The zones, racks, and server structure used by the Knowledge Graph is invented rather than sourced from an operating facility, so results demonstrate the reasoning approach rather than validated performance on a real layout.
Latency vs. reasoning-depth trade-off	The pipeline chains multiple stages — event ingestion, detection models, graph updates, RAG retrieval, and multi-agent reasoning — creating an inherent tension between reasoning depth and response speed that is acceptable in a research prototype but would need to be addressed for production use.
Evaluation ceiling	Because cross-domain incidents are synthetically injected rather than naturally occurring, correlation-accuracy metrics reflect performance on designed scenarios rather than unseen, real-world incident patterns.

DC-Guardian's scope is bound to demonstrate cross-domain correlation and knowledge-grounded incident reasoning across physical security, environmental monitoring, and cybersecurity data, using a synthetic but representative data center topology. The project deliberately excludes live hardware deployment, novel model development, and production-grade performance guarantees, positioning it as a research-oriented proof-of-concept rather than a commercial-ready product.

Data Collection and Preparation

DC-Guardian adopts a hybrid multimodal data collection strategy because no single public benchmark dataset provides the synchronized physical-security, cybersecurity, and equipment-maintenance observations required by the proposed system. The public datasets are used for model development and evaluation, operational documents are used as the RAG knowledge base, and simulated data-center topology and asset metadata supply the relational context required by the Neo4j Knowledge Graph.

The table below summarizes the data and knowledge sources used for each domain:

Table 5 - Data Collection by Domain

Domain	**Source**	**Planned Model / Method**	**Model Output**
Hardware & Predictive Maintenance	AI4I 2020 Predictive Maintenance Dataset	Train and compare suitable anomaly/failure-detection models, including Isolation Forest as a baseline.	Normal/anomalous state, failure risk or anomaly score
Cybersecurity & System logs	Loghub – OpenSSH server logs	Train/compare suitable SSH/log anomaly-detection approaches	Normal/anomalous SSH event and anomaly score
PPE Compliance	SH17 PPE dataset	Fine-tuned YOLOv8 object detector	Detected person/PPE items and compliance status
Face Access Verification	Facial recognition image dataset	OpenCV-based face detection/recognition pipeline	Registered/authorized person or unknown person
Operational Knowledge	SOPs, compliance manuals and response procedures	Chunking + embeddings + vector-backed RAG	Retrieved operational evidence and recommended procedure

Infrastructure Context	Simulated data center topology and asset metadata	Neo4j Knowledge Graph	Relationships among zones, racks, servers, people and events

Each domain will undergo appropriate cleaning, feature preparation and training, validation, and test splitting. After model selection, inference outputs will be normalized into a common event schema containing fields such as timestamp, event type, source, person ID, zone/rack/server ID, model name, prediction, confidence or anomaly score, and severity.

The main challenge in our project is that the source datasets are independent. To avoid presenting artificially created correlations as real observations, DC-Guardian explicitly separates the data into three categories.

1- Original Public training/ evaluation data.
2- Synthetic temporal/ spatial alignment and injected incident scenarios.
3- Live or replayed model inference during the demonstration.

A Python-based event producer will replay prepared telemetry and SSH records at configurable intervals and process camera frames through the PPE and face/access models. The standardized events will then be linked to Neo4j and used to retrieve relevant SOP passages through RAG, enabling the later Agentic AI layer to reason over correlated events.

The table below summarizes which project data is public/real, and which is synthetically created for infrastructure context and scenario simulation:

Table 6 - Data Types: Public/Real and Synthetic

Data	**Real/Public**	**Synthetic**	**Purpose**
AI4I telemetry	✓		Train/evaluate maintenance model
OpenSSH logs	✓		Train/evaluate cyber model
SH17 PPE	✓		Train/fine-tune PPE model
Face images	✓		Develop face/access model
SOP documents		✓	RAG knowledge
DCIM schema concepts		✓	Guide to topology design
Zones/racks/servers		✓	Neo4j infrastructure
Person authorization		✓	Neo4j access context
Dataset-to-server mapping		✓	Simulation

Cross-domain incident timing		✓	Scenario simulation

Infrastructure Context Note: the data center topology is intentionally synthetic and will represent zones, racks, servers, cameras, access points, cooling assets, personnel authorization, and event-to-asset mappings. Public dataset/model outputs will be mapped to these controlled entities during simulation. They will not be presented as observations originally collected from the same physical data center.

Proposed System Design

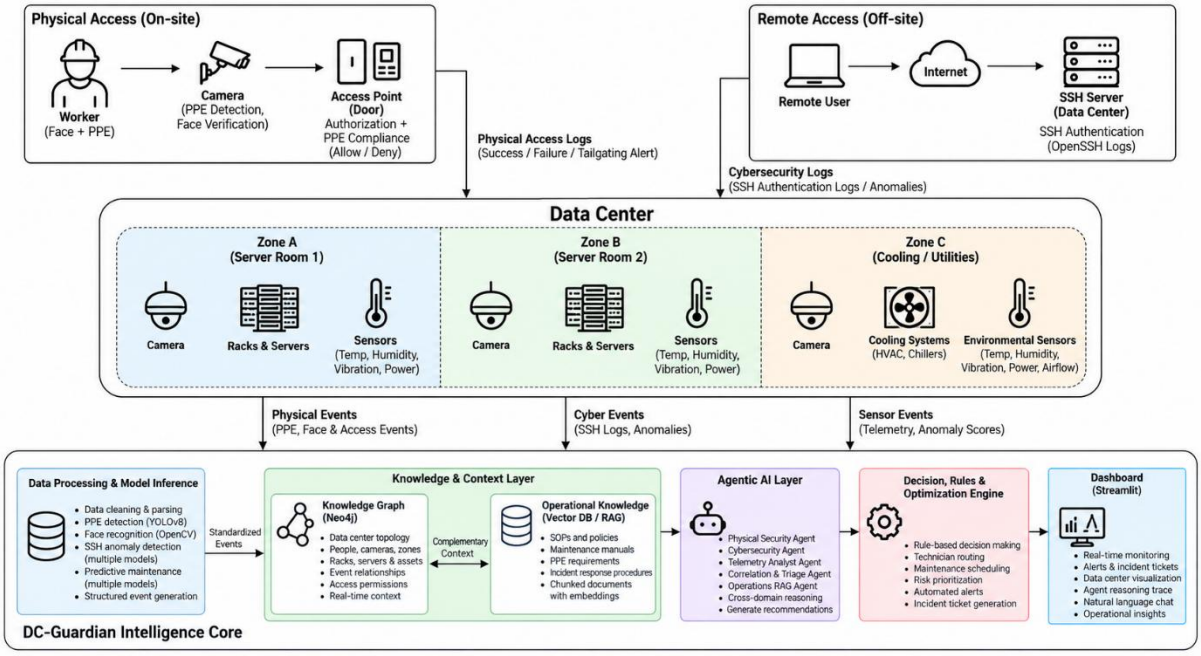


Figure 1 - Proposed System Design [Generated by ChatGPT Image Generation]

DC-Guardian is designed as an intelligent monitoring system for a single data center environment, as shown in figure 1, the proposed system combines three main sources of information: physical security events, cybersecurity events, and equipment/sensors telemetry. These events are fed into the DC-Guardian Intelligence Core to identify anomalies, establish relationships between events, and provide actionable recommendations to data-center operators.

Data Center Environment

The simulated data center environment is divided into multiple zones containing servers, racks, cameras, environmental sensors, and cooling equipment. The environment as well supports three main communication and monitoring capabilities:

- Physical security: Cameras and access points monitor personnel entering the data center. Face recognition and PPE detection are used to determine personnel identity, access authorization, and PPE compliance.
- Cybersecurity: Remote users access data center servers through SSH. Authentication logs are analyzed to identify unusual or potentially malicious login activity.
- Equipment Monitoring: Sensors monitor equipment and environmental conditions such as temperature, humidity, vibration, and power usage.

These three domains generate events that are then forwarded to the DC-Guardian Intelligence Core.

Data Processing and Model Inference

The incoming data is cleaned, parsed and processed by the appropriate machine-learning or deep-learning models. The system uses computer vision models for PPE and face/access verification anomaly-detection models for SSH activity and predictive-maintenance models for equipment telemetry.

The outputs are converted into a common event format containing fields such as timestamp, event type, person or asset identifier, data-center zone, model prediction, confidence or anomaly score, and severity. This enables information from different sources to be processed consistently.

Knowledge and Context

DC-Guardian uses two complementary knowledge components. The Neo4j Knowledge Graph represents the data-center topology and relationships between entities such as people, cameras, zones, racks, servers, sensors, and events. It allows the system to determine whether events from different domains are related. For example, unauthorized personnel detected in Zone A may be correlated with SSH anomaly affecting a server located in the same zone.

The second component is the RAG knowledge base, which contains SOPs, security policies, PPE requirements, maintenance manuals, and incident-response procedures. Relevant information is retrieved when an incident occurs to provide operational guidance.

Agentic AI

The Agentic AI layer analyzes the model output together with information from the Knowledge Graph and RAG knowledge base. Specialized agents evaluate physical-security, cybersecurity, and equipment events, while a correlation agent determines whether multiple events may represent a single incident. The agents then produce an incident assessment and recommend a response based on the available evidence.

Decision, Rules and Optimization

The proposed system includes rules and optimization components to support operational decision-making. Rules can be used to determine the severity and escalation requirements,

while optimization can support activities such as technician assignment and maintenance scheduling.

Dashboard

The results are displayed through the Streamlit dashboard. The dashboard provides operators with real-time monitoring, alerts, incident tickets, data-center topology visualization, agent recommendations, reasoning information, and a natural-language chat interface.

The overall system flow can be summarized as below Figure 2.

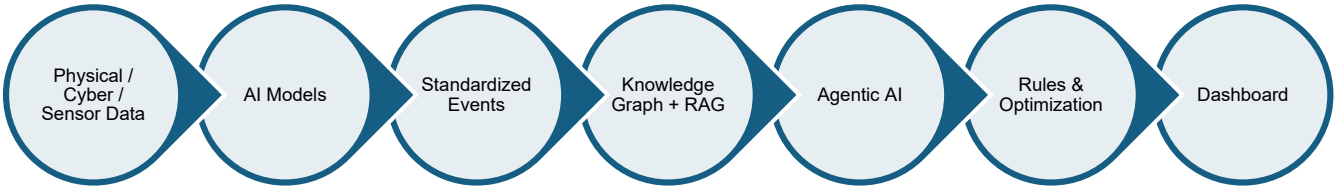


Figure 2 - System Flow

Generative AI Use Disclosure

Tool: ChatGPT (GPT-5.6) and ChatGPT Image Generation

Purpose of use: Generative AI was used throughout the development and refinement of the DC-Guardian project proposal to support market and literature research, identification of relevant research areas and potential academic sources, drafting and structuring of project content, language refinement, and clarification of technical concepts. It was also used to support the development of the Data Collection and Preparation strategy, conceptual refinement and visualization of the proposed System Design, generation and refinement of the system architecture diagram, and review of the overall proposal for clarity, grammatical accuracy, terminology consistency, internal consistency, and presentation. All AI-generated content, suggested references, architectural concepts, and proposed modifications were reviewed and validated by the project team before being incorporated into the proposal. Generative AI was used as a supporting tool and not as a substitute for the project team's validation of the proposed system, datasets, references, technical decisions, or final submitted content.

Prompts used:

"Give me an overview of the market landscape regarding the use of data center protection and maintenance monitoring intelligence systems. Provide research papers on market trends related to cyber-physical security and AI-powered predictive maintenance platforms."

"I am preparing for the dataset collection. Assist me in what I should write. I want first to train multiple models, then simulate a real-time flow of data such as a person's face and PPE

compliance, cybersecurity SSH anomalies, and use Knowledge Graph, Agentic AI and RAG. ***[Provided a document based on what I have started working on with the list of datasets.]****"*

"Does this diagram reflect our idea, or does it need more enhancement? I want one data center with different points of communication, either physical or cyber, and the sensors within the data center. Can you generate the image diagram for me after modification and incorporate it into my diagram style? ***[Provide a draft of the diagram made by draw.io.]****"*

"This is considered the proposed system design. Can you give detailed information related to it? I want a simple system design."

"This is our project proposal document. What do you think about it?"

"Can you tell me where exactly I should modify?"

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